Social Media Signals and Prediction Market Outcomes: A Data-Driven Study on Polymarket

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1 Introduction

Prediction markets have emerged as powerful instruments for aggregating collective intelligence about the likelihood of future events. Among these, Polymarket[1] has established itself as one of the most prominent decentralized platforms, allowing users to trade on the outcomes of real-world events spanning politics, economics, science, and sports. The prices observed on Polymarket reflect the market's aggregate belief about the probability of each outcome, making them a valuable and continuously updated ground truth for forecasting research [1, 2].

At the same time, social media platforms such as Reddit, X (formerly Twitter), and Telegram have become primary arenas where public opinion forms, narratives spread, and collective sentiment consolidates around events of common interest. The discourse unfolding on these platforms through posts, threads, comments, and reactions encodes rich signals about how communities perceive and anticipate real-world developments [3, 4]. Whether and to what extent these signals are informative about the future resolution of prediction market contracts is an open and scientifically relevant question.

The general objective of this challenge is to study this relationship empirically. Specifically, we consider Polymarket as the source of ground-truth outcome data, and three major social media platforms (**Reddit**, **X (Twitter)**, and **Telegram**) as sources of public discourse. The thematic focus of the analysis spans three domains in which Polymarket is particularly active: **political events**, **financial and macroeconomic outcomes**, and **sports and entertainment**. The aim of the challenge is threefold: *i)* to build a structured dataset linking social media discourse to Polymarket contracts and their resolution outcomes through a systematic ingestion pipeline; *ii)* to perform descriptive and analytical tasks that characterize the relationship between social signals and market outcomes across platforms and thematic domains; and, optionally, *iii)* to exploit the collected social media features to train a predictive model for contract outcome forecasting.

2 Data

The dataset to be constructed comprises two interconnected components: **Polymarket contract data** and **social media discourse data**. Both components must be collected, linked, and organized into a unified schema.

Polymarket Contracts

For each market contract considered in the study, the following fields should be collected:

- `market_id`: unique identifier of the contract as assigned by Polymarket.

[1] `https://polymarket.com`

- `question`: the full text of the question posed by the market (e.g., *"Will candidate X win the 2024 presidential election?"*).
- `category`: thematic domain of the contract (`politics`, `finance`, or `sports`).
- `outcomes`: list of possible outcomes available for trading (e.g., `Yes`/`No`, or a set of named candidates).
- `resolution_outcome`: the outcome that was declared correct upon market resolution.
- `resolution_date`: date on which the market was resolved.
- `creation_date`: date on which the market was created and opened for trading.
- `close_date`: date on which trading was closed.
- `url`: canonical URL of the market on Polymarket.

Polymarket Price Time Series

For each contract, the historical evolution of market prices must be collected at regular time intervals (e.g., hourly or daily snapshots), capturing how the market's implied probability of each outcome changed over time. Each snapshot should include:

- `market_id`: reference to the parent contract.
- `outcome`: the specific outcome this price refers to.
- `timestamp`: date and time of the snapshot.
- `price`: market price at the snapshot, interpretable as the implied probability of that outcome (range: 0–1).
- `volume`: cumulative trading volume at the snapshot, where available.

Social Media Post Identification and Provenance

For each social media post or thread collected, the following fields should be gathered:

- `post_id`: unique identifier of the post as assigned by the source platform.
- `platform`: the platform from which the post was collected (`reddit`, `twitter`, or `telegram`).
- `url`: canonical URL of the post or thread.
- `market_id`: reference to the Polymarket contract this post is associated with. The linking can be established through traditional techniques such as keyword matching or entity-based retrieval, or through LLM-based pipelines that leverage the semantic understanding of the contract question to identify relevant posts in a zero-shot or few-shot setting.
- `search_query`: the query or keyword used to retrieve the post during collection.
- `collected_at`: timestamp indicating when the post was retrieved.
- `category`: thematic domain inherited from the associated Polymarket contract.

Author Information

- `author_id`: unique identifier of the account that published the post.
- `author_name`: display name of the author.
- `author_url`: URL of the author's profile page on the platform.
- `author_followers`: number of followers or subscribers of the author at the time of collection, where available.

Post Metadata and Textual Content

- `published_at`: date and time at which the post was published.
- `text`: full textual content of the post or thread opening message.
- `hashtags`: list of hashtags included in the post, where applicable.
- `mentioned_entities`: named entities mentioned in the post (persons, organizations, locations), extracted via NER.
- `language`: detected language of the post.

Engagement Metrics

- `like_count`: number of likes or upvotes at the time of collection.
- `reply_count`: number of replies or comments.
- `repost_count`: number of retweets, shares, or reposts, where available.
- `view_count`: number of views or impressions, where available.

Comments and Replies

- `comments`: a sample of replies or comments associated with each post. Each comment should include, where available, the comment text, the author identifier, the publication timestamp, and per-comment engagement metrics.

Each group should collect data covering at least 100 resolved Polymarket contracts per thematic domain, with associated social media posts spanning the full lifetime of each contract from creation to resolution.

3 Methodology

The project, illustrated in Figure 1, is structured around two main tasks: the construction of the ingestion pipeline and the analytical study of the collected data. Groups may optionally undertake a third task focused on outcome prediction.

Task 1 — Data Ingestion Pipeline

Each group must design and implement a reproducible pipeline for collecting, linking, and preprocessing Polymarket and social media data. The pipeline should handle:

1. **Polymarket data collection:** retrieval of resolved market contracts and their full price time series using the Polymarket API[2]. Groups should target contracts that have already been resolved, so that the ground-truth outcome is available for downstream analyses.

2. **Social media data collection:** systematic retrieval of posts, threads, and comments related to each selected Polymarket contract from the three platforms. Data acquisition methods include API integration (e.g., Reddit API via PRAW) and compliant web scraping where APIs are unavailable or insufficient. Posts should be collected longitudinally, covering the entire period between contract creation and resolution.

3. **Contract-to-post linking:** construction of a mapping between each Polymarket contract and its associated social media posts. This can be achieved through traditional techniques such as keyword extraction from the contract question, entity-based matching, or hashtag tracking, as well as through LLM-based pipelines that exploit the semantic content of the contract question to retrieve and filter relevant posts in a zero-shot or few-shot setting. Groups should document and justify their linking strategy, and report coverage statistics (e.g., fraction of contracts with at least k associated posts).

4. **Preprocessing and normalization:** cleaning of textual fields, handling of missing values, deduplication, language detection, and normalization of timestamps to a common timezone. Named entity recognition (NER) should be applied to extract structured references to persons, organizations, and locations from post text.

5. **Storage:** organization of the collected data into a structured, queryable format suitable for the analytical tasks described below. Each group must choose an appropriate storage solution and **motivate their choice**, discussing why the selected format is well-suited to the characteristics of the dataset, including its temporal structure, the heterogeneity of social media schemas across platforms, and the query patterns required by the analyses.

Task 2 — Analytical Dashboard

Using the constructed dataset, each group must perform a set of descriptive and analytical studies. The following analyses are required as a minimum; each group is encouraged to extend them with additional investigations.

1. **Descriptive analysis of Polymarket contracts:** characterize the collected contracts along basic dimensions, including distribution by thematic domain, contract duration, trading volume, and price volatility. Analyze how market-implied probabilities evolve over the lifetime of a contract, and whether prices at different temporal distances from resolution are informative of the final outcome.

2. **Social media discourse analysis:** analyze the volume, timing, sentiment, and linguistic content of posts associated with each contract. Techniques such as sentiment analysis, topic modeling, and named entity frequency are suggested. Compare discourse patterns across platforms and thematic domains.

3. **Signal–market correlation analysis:** study the temporal relationship between social media signals (post volume, aggregate sentiment, engagement) and Polymarket price movements. Identify whether spikes in social activity precede, coincide with, or follow significant

[2] `https://docs.polymarket.com`

price shifts, and whether the direction of sentiment is aligned with the direction of price change. Of particular interest is the analysis of whether changes in social media activity such as sudden increases in posting volume or rapid shifts in aggregate sentiment are temporally correlated with significant movements in market-implied probabilities. Such signals can be extracted through traditional time-series methods as well as through LLM-based pipelines capable of detecting nuanced semantic shifts in discourse that may not be captured by lexicon-based sentiment tools.

4. **Cross-platform comparison:** analyze whether different platforms carry complementary or redundant signals with respect to market outcomes. Identify contracts for which one platform is substantially more informative than the others, and characterize what distinguishes such cases.

Each group must develop an interactive visualization platform (e.g., using Streamlit [5]) that exposes the results of the analytical study. The dashboard should allow users to:

- browse the collected Polymarket contracts by thematic domain, resolution date, and outcome;
- visualize the price time series of a contract alongside the temporal evolution of associated social media activity;
- explore sentiment and engagement distributions per platform and domain;
- inspect individual posts linked to a contract and examine their metadata and extracted features.

Task 3 (Optional) — Outcome Prediction from Social Media Features

This optional task focuses on exploiting the social media signals collected in Task 1 to train a predictive model for Polymarket contract outcomes. The core research question is whether the discourse observed on social media platforms — independently of, or in combination with, the market price itself — carries statistically useful information for forecasting the resolution outcome of a contract.

Feature engineering. Starting from the raw social media data, suggested feature categories include (but are not limited to):

- **Volume features:** total number of posts, comments, and temporal growth rate of posting activity.
- **Engagement features:** aggregate like, reply, repost, and view counts; ratio of high-engagement to low-engagement posts.
- **Sentiment features:** average and variance of sentiment scores computed over post texts; fraction of positive, negative, and neutral posts; sentiment trend over time within the window.
- **Linguistic features:** TF-IDF or embedding-based representations of the most frequent terms; presence of specific named entities or keywords associated with the contract outcomes.

Prediction task and models. The prediction target is the **resolution outcome** of the contract, treated as a binary or multi-class classification problem depending on the number of available outcomes. Models should be evaluated using standard classification metrics (accuracy, macro-F1, AUC-ROC) under a proper temporal cross-validation scheme that respects the chronological ordering of contracts and prevents data leakage.

4 Deliverables

Each group is expected to submit the following materials upon project completion.

1. **Technical report**: a document describing all methodological choices made throughout the project, including the data collection strategy, the contract-to-post linking approach, the preprocessing steps, the storage solution and its motivation, and the analytical findings. Groups undertaking Task 3 should include a dedicated chapter reporting the predictive model design, the feature engineering choices, the experimental setup, and the results in terms of standard classification metrics (accuracy, macro-F1, AUC-ROC).
2. **Dataset**: the full structured dataset produced by the ingestion pipeline, organized according to the schema described in Section 2 and stored in the chosen format.
3. **Source code**: all code produced during the project. This includes the data ingestion pipeline, the preprocessing scripts, the analytical notebooks, and the dashboard application. Groups undertaking Task 3 should also include the code for feature engineering, model training, and evaluation.
4. **Dashboard**: a running instance of the interactive visualization platform described in Task 2, either deployed online or executable locally following the instructions provided in the source code repository. **The dashboard will be demonstrated live during the oral discussion.**
5. **Slide Presentation** (optional): a slide deck summarizing the project, covering the motivation, the data collection approach, the main analytical findings, and, if applicable, the outcome prediction results. The presentation should be concise and oriented toward a technical audience.

5 Resources

Beyond all technologies presented during the course, the following additional libraries and tools may be useful:

- **Polymarket API**[3]: official API for retrieving market contracts, price history, and trading data from Polymarket.
- **PRAW**[4]: Python Reddit API Wrapper, suitable for collecting posts, comments, and subreddit activity from Reddit.
- **HuggingFace**[5]: provides a wide range of pre-trained language models for tasks such as sentiment analysis, stance detection, named entity recognition, and text embedding.
- **Streamlit**[6]: framework for building and deploying interactive data applications with customized user interfaces.
- **Apache Spark / PySpark**: for scalable preprocessing and analysis of large-scale longitudinal datasets.
- **Scrapling**[7]: an adaptive web scraping framework.

The official **Teams** channel represents the forum to ask questions and share doubts throughout the project.

[3] `https://docs.polymarket.com`
[4] `https://praw.readthedocs.io`
[5] `https://huggingface.co`
[6] `https://streamlit.io`
[7] `https://github.com/D4Vinci/Scrapling`

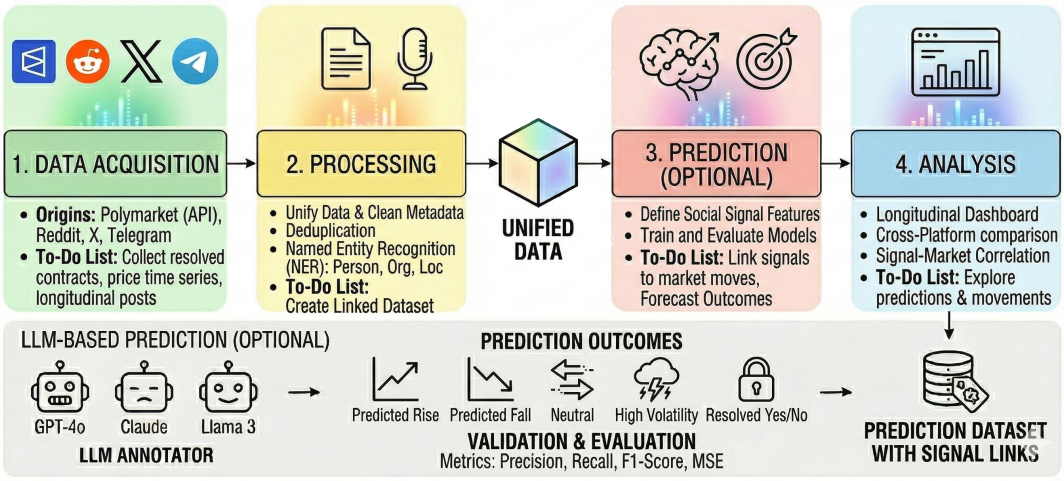


Figure 1: An example of the Methodology pipeline

6 Evaluation

The project will be evaluated by a committee. Each member of the committee will assign a score from 0 to 5 to several aspects related to the methodology and the final dashboard. Scores will then be averaged. The evaluation of the methodological aspects can slightly change as a result of the oral discussion and will not be revealed before the end of the oral discussion.

Methodology aspects

- *Design choices*: Technologies are properly chosen and discussed, including the storage solution, the contract-to-post linking strategy, and the feature engineering choices for the analyses;
- *Experiments*: Quality of the proposed experimental evaluation, including the rigor of the correlation analyses and, for groups undertaking Task 3, the soundness of the predictive modeling and ablation study;
- *Creativity*: The ability to adopt innovative and creative approaches in solving problems and addressing challenges encountered.

Dashboard aspects

- *Priority*: The dashboard should accurately prioritize the main ideas of the project;
- *Accuracy*: The dashboard should not omit any important details;
- *Conciseness*: The dashboard should convey the most important information in a brief and to-the-point manner, avoiding unnecessary repetition or elaboration;
- *Usability*: The dashboard should be insightful and easy to use by the end user;
- *Customization*: The dashboard could offer customization features to allow users to tailor visualizations and analyses to their specific needs and interests.

References

[1] J. Wolfers and E. Zitzewitz. Prediction markets. *Journal of Economic Perspectives*, 18(2):107–126, 2004.

[2] J. Berg, F. Nelson, and T. Rietz. Prediction market accuracy in the long run. *International Journal of Forecasting*, 24(2):285–300, 2008.

[3] J. Bollen, H. Mao, and X. Zeng. Twitter mood predicts the stock market. *Journal of Computational Science*, 2(1):1–8, 2011.

[4] M. Salathé and S. Khandelwal. Assessing vaccination sentiments with online social media: implications for infectious disease dynamics and control. *PLOS Computational Biology*, 8(10):e1002199, 2012.

[5] Streamlit Inc. Streamlit: The fastest way to build data apps. `https://streamlit.io`, 2023.